IC Memo - Zeckathon PS-01: India capital-market infrastructure | Recommendation: CAMS (relative winner) - conditional accumulation, INR 0 deployed at CMP

Price date 2026-10-29: Yahoo NSE daily bar / last trade at 15:29 IST (not cross-checked to the NSE bhavcopy). All figures INR; generated by run_all.py from cached, sourced data (data/sources.csv). Assumptions labelled in config/assumptions.yaml.

1. Recommendation

CAMS is the best of the four on every model metric: highest probability-weighted 12-month return, smallest gap between CAPM Ke (13.2%) and market-implied Ke (11.2%), and zero direct exposure to the derivatives-turnover shock. **12-month target price INR 539** (base DCF rolled forward at Ke, net of dividends) vs CMP INR 696.95: **-23%**; bear INR 340 (-51%), bull INR 705 (+1%). Honest read: at today's price none of the four clears our cost of equity; the market is pricing CAMS at ~11.2% Ke. For a long-only mandate this means: CAMS is the stock to own in this group, but **not at today's price**. INR 500 cr is committed only through model-derived entry bands (Section 5); at CMP nothing is bought. Expected deployment is INR 139 cr by 6 months and INR 208 cr by 12 months (volatility model; INR 145 cr on CAMS's own 5y history); the rest stays in liquid funds and is released back to the fund if bands are not reached by the 12-month review.

2. Comparable valuation (identical engine, drivers and CAPM method for all four)

	CMP	Beta	Ke	DCF/sh	TP base	vs CMP	TP bear	TP bull	PW ret	Impl. Ke	P/E 28E	EV/EB 28E	ROCE 26	CFO/PAT
CAMS	697	1.24	13.2%	490	539	-23%	340	705	-21%	11.2%	29.5x	19.6x	47%	1.14x
KFINTECH	860	1.31	13.6%	506	560	-35%	306	771	-34%	10.9%	32.1x	21.8x	29%	1.15x
BSE	3502	1.37	13.9%	1347	1503	-57%	1046	1869	-57%	9.1%	43.9x	33.3x	60%	1.34x
CDSL	1282	1.63	15.2%	321	358	-72%	237	456	-72%	8.5%	56.5x	40.8x	32%	1.00x

Ke = (10y G-sec 7.40% - India CDS 0.66%) + Blume beta (3y weekly vs Nifty 50) x ERP 5.23% (Damodaran Jan-2026, CDS-based); WACC = Ke (net-cash). FCFF: FY27E (reported YTD + run-rate) to FY31E explicit, 10-year linear fade to g = 6.5%, Gordon terminal. Equity = EV + owned net cash (clearing/SGF/trust funds excluded) + associates (BSE's 15% CDSL stake at market) - NCI - debt-like - dividends paid since balance-sheet date. Ke decision (cross-examined, not tuned): CAMS's regression beta 95% CI is 0.99-1.49; the beta its price implies is 0.86 (outside that CI). Damodaran India industry betas inside the CI: 'Information Services' 1.02 -> TP INR 646 (CMP 697). Base case keeps the regression beta (Appendix B3). PW = 25/50/25 bear/base/bull. P/E and EV/EBITDA are cross-checks only; FY28E P/E at CMP: CAMS 29x, KFin 32x, BSE 44x, CDSL 57x.

3. Why the other three are less attractive

BSE: best franchise momentum (ROCE 60%) but ~65% of FY26 revenue is equity derivatives; the 30% turnover stress alone cuts the TP to INR 1339. Implied Ke 9.1% means the price already assumes the post-SEBI/RBI/STT regime normalises; CEO retires Jun-2027. **CDSL**: richest multiple (P/E FY28E 57x) on falling FY26 EPS; implied Ke only 8.5% -> worst risk/reward. **KFin**: faster growth (21% EPS CAGR FY27-29E) but ranks #2 because of higher debtor days (76), acquisition goodwill, an unverified INR 676 cr earn-out-type liability and a General Atlantic supply overhang (14.06% left); bear TP -64% vs CMP.

4. Moat, financial quality, governance

CAMS: 67.2% MF AAUM share, multi-year AMC contracts, high switching costs; ROCE 47%, CFO/PAT 1.14x (FY24-26), no promoter, related-party dealings limited to KMP pay (AR Note 30). Flags: intangibles under development up from INR 38 cr to INR 88 cr and INR 123 cr capital commitments for the new RTA platform (capitalisation and migration risk); TER-to-BER regime (Apr-2026) keeps yield pressure on. Integrated 3-statement model: balance sheet balances every year FY27-31E (max |check| 1.1e-12), CF closing cash = BS treasury, FCFF identical to DCF engine.

CAMS 3-statement (INR cr)	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue	1580	1788	2026	2297	2608
EBITDA	731	847	982	1139	1323
PAT	508	591	687	798	929
EPS (INR)	20.4	23.7	27.5	32.0	37.3
CFO	607	692	800	927	1075
FCFF (3S = DCF engine)	427	487	569	664	775
Closing treasury (CF = BS)	965	1084	1224	1388	1580
Total assets	1973	2185	2430	2714	3044
Assets - liabilities - equity	-0.00	-0.00	-0.00	-0.00	-0.00

5. Deploying INR 500 cr (~2.9% of CAMS market cap)

Tranche	INR cr	Buy at/below	Fall vs CMP	P(touch) 6m	P(touch) 12m	Hist. 12m	Days to fill	Band basis
T1 starter	100	637	-9%	67%	76%	70%	5	bull value today; after Q2 FY27 results
T2	150	564	-19%	32%	48%	34%	7	mid of bull and base value
T3	250	490	-30%	10%	24%	9%	11	base value today

P(touch) = probability the price trades at or below the band within the horizon: driftless log-normal, sigma 30% (252-day); 'Hist.' = share of all 12-month windows in CAMS's 2021-2026 daily history that touched the same % fall. Liquidity: ADV INR 109 cr (NSE+BSE, 60 sessions); at 20% participation the full INR 500 cr needs ~23 sessions once bands are hit. **Fallback**: undeployed capital sits in overnight/liquid funds (not an index proxy - every listed peer is above our value); hard review at 6 months (after Q4 FY27 results): bands are re-computed from the re-run model, never lowered or raised by hand; capital still undeployed at 12 months is returned to the fund rather than forced into the stock. Buying a starter at CMP is rejected: probability-weighted 12m total return at CMP -21%.

Catalysts: Q2 FY27 results / analyst meets (Nov-2026); SIP and equity-AAUM growth; non-MF (KRA, insurance repository, AIF/PMS) scaling at 20%; go-live of the re-architected RTA platform; any easing of AMC fee pressure.

Risks: equity drawdown (asset-based fees 74% of revenue); AMC yield cuts under BER; client migration to KFin; platform-migration cost overruns; a market de-rating of the whole sector (our Ke already exceeds the implied one).

Exit / revisit triggers: (1) MF AAUM market share below 65% for two consecutive quarters or loss of a top-10 AMC -> exit; (2) asset-based revenue growth more than 5pp below AAUM growth for two quarters (yield compression) or EBITDA margin below 42% -> revisit/trim; (3) price above bull TP INR 705 without bull-case drivers visible -> trim to zero.

6. Bear case (brief-mandated) and isolated stresses

Scenario	Ticker	Rev FY28E	EBITDA m FY28E	EPS FY28E	TP	vs CMP
base	CAMS	1788	47.4%	23.6	539	-23%
bear	CAMS	1637	43.8%	20.0	340	-51%
deriv_only	CAMS	1788	47.4%	23.6	539	-23%
equity_only	CAMS	1682	44.9%	21.1	482	-31%
base	KFINTECH	1757	39.8%	26.8	560	-35%
bear	KFINTECH	1618	35.4%	21.8	306	-64%
deriv_only	KFINTECH	1757	39.8%	26.8	560	-35%
equity_only	KFINTECH	1661	36.9%	23.3	486	-44%
base	BSE	6220	65.3%	79.8	1503	-57%

bear	BSE	5342	62.9%	67.2	1046	-70%
deriv_only	BSE	5594	63.7%	70.8	1339	-62%
equity_only	BSE	6117	65.1%	78.3	1477	-58%
base	CDSL	1308	47.8%	22.7	358	-72%
bear	CDSL	1199	43.5%	19.1	237	-81%
deriv_only	CDSL	1304	47.6%	22.5	356	-72%
equity_only	CDSL	1236	45.0%	20.3	321	-75%

Bear = derivatives turnover -30% AND equity market -20% as persistent level shifts from Q3 FY27, segment growth -3pp FY28-31, g -1pp. Each segment carries an explicit elasticity to derivatives turnover and to market level (config/assumptions.yaml); fixed costs do not flex, so margins compress.

7. Sensitivity discipline (CAMS 12m TP, INR; every cell a full model re-run)

WACC \ g	5.0%	5.5%	6.0%	6.5%	7.0%	7.5%
11.2%	624	661	705	759	825	908
12.2%	539	565	595	630	671	722
13.2%	475	494	515	539	567	600
14.2%	426	440	455	473	493	516
15.2%	386	397	409	422	437	453

Driver (low / high)	TP low	TP high	Range
Growth FY28-31 -/+3pp	416	690	274
Beta +/-0.2	470	634	164
Ke +/-100bp	473	630	157
Fixed-cost growth 13%/7%	492	583	92
Equity market -20%/+10%	482	568	86
Capex % rev +2pp/-2pp	503	575	72

Also in outputs/: ERP x beta, AAUM-growth x market-shock (BSE: ADTV x realisation) grids for all four, scenario-weight sensitivity. Reading: CAMS TP exceeds CMP in 5 of 30 WACC/g cells, all at WACC 11.2% - the price embeds a low discount rate.

8. Data gaps / unresolved (marked, not estimated silently)

Q2 FY27 results for BSE/CDSL/CAMS not yet released (run-rates use Q1); CAMS FY26 segment split approximated from Q1/Q4 FY26 mix; KFin variable-cost share assumed 10%; KFin INR 676 cr non-current financial liability composition unverified; CDSL other current financial liabilities deducted conservatively; dividend payment dates assumed before price date; BSE Q2 ADTV is broker (HSIE) data; ADV from Yahoo (NSE+BSE), not exchange bhavcopies.

Appendix A - Industry: India's capital-market 'toll collectors'

A1. Value chain, licences and market structure. Every cell restates a cached source (IDs -> data/sources.csv; local copies in data/raw/).

	Layer	Licence / regulator	Structure & share	Revenue model	Moat / switching cost	Threats	Sources
BSE	Exchange	Recognised stock exchange + clearing corp (ICCL); SEBI-regulated MII	Duopoly with NSE in equity derivatives; since 20-Nov-2024 each exchange may offer weekly expiry on only one benchmark index (BSE: Sensex)	Transaction charges (options on premium; options ~95% of txn charges), listing fees, data, StAR MF	Liquidity network effect; Sensex weekly-expiry franchise; listing base	Regulatory volume curbs, STT hike (1-Apr-2026), RBI prop-funding rules (1-Jul-2026), closing-auction change, NSE pricing response	SEBI_CIRC_132, MINT_CAS, BS_STT, MC_RBI, BSE_PPT_Q1
CDSL	Depository	Depository under Depositories Act 1996; SEBI-regulated MII	Duopoly with NSDL; CDSL 18.59 cr demat accounts (30-Jun-2026), ~80% of accounts; incremental account share down ~420 bp since Mar-26 to 81.4%	Annual issuer charges (per folio/ISIN), debit-transaction charges, IPO/corporate actions, KYC/e-voting/data	Mandatory dematerialisation; DP and issuer connectivity; annuity-like issuer fees	True-to-label uniform pricing; fintech brokers routing new accounts to NSDL; market-cyclical transaction/IPO fees	CDSL_TR_Q1, SEBI_CIRC_92, CDSL_PPT_Q1
CAMS	MF RTA	SEBI-registered Registrar & Transfer Agent; also KRA, insurance repository	Two-player MF RTA market: CAMS 67.2% of MF AAUM (Q1 FY27, ex-FoF basis); industry AAUM INR 83.1 tn; CAMS live-SIP share 63.9%	bps of serviced AAUM (asset-based ~74% of revenue) + per-transaction fees + non-MF platforms	AMC contracts, investor-record migration cost, 23-year average relationship with top-10 MF clients	AMC fee renegotiation under TER->BER (1-Apr-2026); share loss to KFin; platform re-architecture execution	CAMS_PPT_Q1, CAFE_TER
KFINTECH	MF RTA / issuer registry	SEBI-registered RTA; NPS CRA; international fund admin (Ascent)	#2 MF RTA: 33.1% overall AAUM share, 32.8% equity AAUM, 36.7% of monthly SIP inflows; 50.2% of NSE-500 issuers (Q2 FY27)	bps of AAUM, issuer folio/corporate-action fees, international/alternates admin	Issuer-registry scale; won 254 of last 435 new AMC launches (counted by KFin)	Same AMC pricing pressure; acquisition integration (Ascent earn-out); promoter sell-down overhang	KFIN_PPT_Q2, KFIN_ASCENT, KFIN_SHP

A2. Concentration (two-player markets).

	Leader	Share %	#2	Share %	HHI	Basis note	Sources
MF RTA (AAUM share)	CAMS	67.2	KFINTECH	33.1	5611	CAMS Q1 FY27 ex-FoF vs KFin Q2 FY27 overall - different quarter/basis, sum 100.3% (not additive by construction)	CAMS_PPT_Q1, KFIN_PPT_Q2
Depository (demat accounts)	CDSL	80.0	NSDL	20.0	6800	CDSL 'approximately 80%' (call); NSDL = remainder (duopoly)	CDSL_TR_Q1

Reading: every layer is a regulated duopoly. Pricing power is capped by SEBI (true-to-label uniform MII charges; TER->BER for AMCs), so moats protect volume share more than price. The exchange's revenue is a toll on trading activity; the depository's is part annuity (issuer fees) and part activity; the RTAs' is a toll on assets (bps of AAUM), which is why the F&O; curbs reach them only indirectly.

A3. Regulatory timeline: SEBI F&O; curbs, fee rationalisation and related shocks

Date	Measure	Who it hits	Sources
01-Jul-2024 (eff. 01-Oct-2024)	True-to-label: MII charges uniform for all members, no volume slabs; end-client charge = amount MII receives	BSE (txn charges), CDSL (DP/transaction charges): pricing reset, lower realisation	SEBI_CIRC_92
01-Oct-2024 (eff. 20-Nov-2024 to 01-Apr-2025)	Index-derivative curbs: upfront premium collection, no calendar-spread benefit on expiry day, intraday position-limit monitoring, contract size INR 15-20 lakh, one weekly expiry per exchange, +2% ELM on expiry day	BSE directly (options premium turnover); CDSL/brokers second-order via fewer active traders	SEBI_CIRC_132
07-Jul-2025	SEBI study: ~91% of individual F&O; traders lost money in FY25; net losses INR 1,05,603 cr (FY24: 74,812 cr); unique individual traders 61.4 lakh (Q1 FY25) -> 42.7 lakh (Q4 FY25); index-options premium turnover -9% YoY (Dec-24 to May-25)	Evidence the curbs bite on participation; supports policy persistence (bear case)	SEBI_PR39
FY26	Index options contracts traded fell ~52% in FY26; NSE/BSE options volumes down >50%	BSE/NSE volumes; CDSL transaction activity	SEBI_AR26, FOW_FO
01-Apr-2026	Budget-2026 STT increase on F&O; SEBI TER -> BER expense framework for MFs	BSE (cost of trading up); CAMS/KFin (AMC economics -> RTA yield pressure)	BS_STT, CAFE_TER
01-Jul-2026	RBI bank-guarantee / proprietary-funding rules; brokers report F&O; volumes -20-25%	BSE premium ADTV (Q2 FY27 QTD ~INR 225 bn/day, -24% QoQ per HSIE)	MC_RBI, HSIE_BSE

A4. F&O; curbs: direct vs second-order effects, quantified with the same engine (12-month TP, INR)

	Deriv.-elastic rev share FY27E	Market-elastic rev share	TP base	TP deriv -30%	Direct TP chg	2nd-order what-if (labelled, magnitude NOT sourced)	TP 2nd-order	2nd-order chg	TP deriv -30% + 2nd	Combined chg
BSE	72%	17%	1503	1339	-10.9%	cash-market & other txn level -10% (fewer active traders)	1493	-0.7%	1329	-11.6%
CDSL	2%	56%	358	356	-0.6%	transaction & KYC/new-account revenue level -10% (fewer active traders/new demats)	345	-3.6%	343	-4.2%
CAMS	0%	60%	539	539	+0.0%	MF asset-based revenue level +5% (household savings redirected from F&O; to MFs)	557	+3.3%	557	+3.3%
KFINTECH	0%	53%	560	560	+0.0%	domestic MF revenue level +5% (same channel as CAMS)	580	+3.6%	580	+3.6%

Direct (revenue that moves one-for-one with derivatives turnover): only BSE - equity derivatives were ~65% of FY26 revenue and options are ~95% of transaction charges (BSE_PPT_Q1, MINT_CAS); the clearing float also scales with margins. A 30% fall costs BSE ~11% of TP even after its variable regulatory/SGF costs fall with revenue. **Second-order**: CDSL loses debit transactions and new-account/KYC fees as active traders fall (unique F&O; traders fell 61.4 -> 42.7 lakh across FY25, SEBI_PR39) but its issuer annuity is untouched; CAMS and KFin have no derivatives revenue - the channel is household savings: if money that left F&O; goes to MFs (CAMS: equity net sales +43% YoY, SIP collections +20.7% YoY in Q1 FY27, CAMS_PPT_Q1), RTA AAUM rises. We found no source that measures that switch, so the +5% is a labelled what-if, not in any scenario. Elasticities e_d / e_m per segment are in config/assumptions.yaml.

Appendix B - Valuation detail (identical engine for all four)

B1. EV -> equity -> 12-month TP bridge (INR cr unless per share). Diluted shares = period-end issued + Ind AS 33 dilutive ESOP increment.

	BSE	CDSL	CAMS	KFINTECH
beta	1.37	1.63	1.24	1.31
ke	13.88%	15.25%	13.24%	13.58%
g	6.50%	6.50%	6.50%	6.50%
pv_explicit	11,110	1,521	2,243	1,604
pv_fade	17,573	2,282	4,230	3,509
pv_terminal	18,415	1,894	5,127	4,004
tv_share	39%	33%	44%	44%
ev	47,098	5,698	11,600	9,117
net_cash	4,811	1,272	777	-336
associates	4,106	48	0	3
nci	122	42	-0	0
dividends_since_bs	412	266	161	0
equity	55,481	6,710	12,215	8,784
shares_issued	41.1979	20.9000	24.7985	17.2872
dilutive_increment	0.0000	0.0000	0.1320	0.0684
shares_diluted	41.1979	20.9000	24.9305	17.3556
value_ps	1346.7	321.0	490.0	506.1
dps12	30.3	11.8	15.5	14.9
target_price	1503.3	358.2	539.3	559.9
cmp	3502.00	1282.00	696.95	859.80
upside	-57.1%	-72.1%	-22.6%	-34.9%

equity = EV + net cash + associates - NCI - dividends paid since the balance-sheet date; TP = value/share x (1 + Ke) - DPS over the next 12 months. An independent reviewer recomputation reproduced all four TPs exactly (pre-dilution fix). Share-count sources: BSE: BSE_AR26: Note 30 EPS: weighted avg shares 41,19,79,077 (AR: basic=diluted, no dilutive instruments); CDSL: CDSL_AR26: Equity share capital 20,900 lakh / FV 10 (AR: basic=diluted EPS, no ESOP); CAMS: CAMS_AR26: = shares_issued_cr + dilutive_increment_cr (period-end issued + ESOP dilution per AR EPS note); KFINTECH: KFIN_PPT_Q2: = shares_issued_cr + dilutive_increment_cr

B2. Relative valuation at CMP (cross-check, not the TP driver)

	P/E TTM	P/E 27E	P/E 28E	EV/EBITDA 26	EV/EBITDA 27E	EV/EBITDA 28E	Rev CAGR 24-26	EPS CAGR 27-29E	ROCE 26 %	EBITDA m 27E	PEG 28
BSE	50.4	49.9	43.9	44.0	37.3	33.3	81%	14%	60	65%	3.2
CDSL	56.9	64.5	56.5	43.8	46.2	40.8	19%	14%	32	47%	4.0
CAMS	34.9	34.4	29.5	24.3	22.7	19.6	15%	17%	47	46%	1.8
KFINTECH	42.6	38.7	32.1	28.8	25.9	21.8	25%	21%	29	38%	1.5

B3. Cost-of-equity cross-examination (why the base Ke was NOT changed)

	Raw beta	SE	Blume beta	CI95 low	CI95 high	Beta implied by CMP	Implied Ke	TP @ CI low	TP base	TP @ CI high	CMP
BSE	1.55	0.32	1.37	0.94	1.79	0.45	9.1%	2025	1503	1227	3502
CDSL	1.93	0.22	1.63	1.34	1.91	0.33	8.5%	418	358	316	1282
CAMS	1.36	0.19	1.24	0.99	1.49	0.86	11.2%	663	539	456	697
KFINTECH	1.46	0.27	1.31	0.96	1.66	0.79	10.9%	774	560	435	860

Damodaran India industry (Jan-2026)	Firms	Levered beta	Unlev. beta (cash-corr.)	BSE TP	CDSL TP	CAMS TP	KFin TP
Brokerage & Investment Banking	186	0.60	0.64	2793	780	994	1154
Information Services	26	1.06	1.02	1894	523	646	723
Business & Consumer Services	75	0.71	0.68	2676	747	949	1098
Computer Services	196	0.73	0.74	2490	694	877	1009
Investments & Asset Management	115	0.51	0.49	3514	986	1274	1498

Decision rule (written after the bottom-up TPs below were computed, so judge it on its logic, not its timing): keep each firm's own 3-year regression beta unless a sourced alternative is consistent with it. Regression betas are precise (SE 0.19-0.32, 156 weeks). Damodaran's India dataset (DAMO_BETA_IN) has no exchange/depository/RTA bucket; the candidate buckets span 0.49-1.02, so picking one would be choosing the answer. For CAMS only 'Information Services' (1.02) is inside the 95% CI; it gives TP INR 646 vs CMP 697. The beta CAMS's price implies (0.86) lies below the CI. ERP alternatives (rating-based 7.08% with matching rf) are in the ERP x beta grid (Appendix D).

Appendix C - Financial due diligence

C1. PAT vs CFO vs FCF and working capital (Screener consolidated FY24-26, SCR_*)

	PAT 24-26	CFO 24-26	FCF 24-26	CFO/PAT	FCF/PAT	CFO/PAT 24	CFO/PAT 25	CFO/PAT 26	Debtor days 26	Debtor days 24	ROCE 24	ROCE 26
BSE	4581	6161	5364	1.34	1.17	3.42	0.31	1.25	39	49	23	60
CDSL	1401	1396	1047	1.00	0.75	0.92	1.03	1.03	21	30	40	32
CAMS	1288	1462	1166	1.14	0.91	1.14	1.03	1.24	14	21	48	47
KFINTECH	923	1058	776	1.15	0.84	1.17	1.20	1.08	76	66	30	29

Reading: BSE's reported CFO includes clearing/settlement-fund movements (CFO/PAT 3.4x in FY24) - only owned cash enters our bridge. CAMS converts profit to cash most reliably (no acquisitions, short receivables); KFin's debtor days and acquisition goodwill make its PAT less cash-backed; CDSL is clean but cyclical.

C2. Governance, related parties, capitalisation and contingent items

	Item	Flag / why it matters	Sources
BSE	MD&CEO; Sundararaman Ramamurthy retires 15-Jun-2027 (age limit); successor search open, applications due 19-Nov-2026	AMBER - key-man/strategy continuity during regulatory transition	BSE_CEO
BSE	Reported operating cash flow includes movements in clearing/settlement funds (CFO/PAT 3.4x FY24; 1.25x FY26)	AMBER - use owned cash only (done in bridge)	SCR_BSE, BSE_AR26
CDSL	BSE holds 15% (promoter); MII governance under SEBI; FY26 EPS decline	GREEN/AMBER	SCR_CDSL, BSE_AR26
CAMS	No promoter; related-party transactions limited to KMP remuneration at arm's length (Note 30); auditor S.R. Batliboi	GREEN	CAMS_AR26
CAMS	Intangibles under development up from INR 38 cr to INR 88 cr; capital commitments INR 123 cr mostly cloud contracts for new RTA platform	AMBER - capitalisation policy and platform migration risk	CAMS_AR26
KFINTECH	Promoter General Atlantic sold 8.75% at INR925 (Aug-2026); promoter holding 14.06% (Sep-26) - overhang	AMBER - supply overhang	KFIN_SHP
KFINTECH	INR 676 cr consolidated non-current financial liability inferred as Ascent 49% earn-out (composition UNVERIFIED); deducted as debt-like	AMBER - UNVERIFIED	KFIN_PPT_Q2, KFIN_ASCENT

C3. Moat and fragility summary

	Moat	Fragility	Rating	Sources
BSE	Exchange duopoly (with NSE); network effects in liquidity; listing/cash franchise	~65% of FY26 revenue = equity derivatives (facts.yaml segments); SEBI F&O; curbs, STT hike (BS_STT), RBI prop-funding rules (MC_RBI) and CAS directly hit volume/realisation	Strong but regulator-dependent	BSE_AR26, BSE_PPT_Q1, SEBI_AR26, MINT_CAS
CDSL	Depository duopoly with NSDL; annual issuer charges are annuity-like	Transaction/IPO/KYC revenue is market-cyclical; FY26 PAT fell YoY (455 vs 526, SCR_CDSL)	Strong, cyclical earnings	CDSL_PPT_Q1, SCR_CDSL
CAMS	RTA duopoly with KFin; 67.2% MF AAUM share (CAMS_PPT_Q1); multi-year AMC contracts, high switching cost (data migration, investor servicing)	Asset-based fees tied to AAUM (54.3% equity) and AMC pricing pressure (TER->BER, CAFE_TER); zero direct F&O; exposure	Strongest; least regulatory-volume exposure	CAMS_PPT_Q1, CAMS_AR26, CAFE_TER
KFINTECH	#2 MF RTA; issuer registry; growing international (Ascent) and alternates	Market-share challenger; acquisitions add goodwill (INR 1,336 cr Sep-26, KFIN_PPT_Q2) and an earn-out (KFIN_ASCENT); debtor days 76 (FY26)	Good, execution/integration risk	KFIN_PPT_Q2, KFIN_ASCENT, SCR_KFIN

C4. CAMS integrated three-statement model (INR cr) - balance sheet, cash flow and DCF tie

Balance sheet	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Net fixed assets	494.2	531.9	574.5	622.9	677.7	739.9
Treasury (cash+inv)	854.4	965.2	1084.1	1223.7	1387.7	1580.3
Receivables	56.6	59.0	66.8	75.7	85.8	97.5
Total assets	1810.4	1973.4	2184.9	2429.9	2714.1	3044.0
Total liabilities	489.5	500.0	534.2	573.3	618.0	669.2
Equity incl. NCI	1320.9	1473.4	1650.6	1856.6	2096.1	2374.8
BS check (A-L-E)	0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Cash flow	**FY27E**	**FY28E**	**FY29E**	**FY30E**	**FY31E**	
PAT	508.5	590.8	686.5	798.3	929.1	
CFO	607.3	691.7	800.5	927.3	1075.1	
Capex	-140.6	-159.1	-180.3	-204.5	-232.2	
Dividends	-355.9	-413.6	-480.6	-558.8	-650.4	
Net change in cash	110.7	118.9	139.6	163.9	192.6	
Closing treasury	965.2	1084.1	1223.7	1387.7	1580.3	
FCFF (3S)	427.1	487.2	568.6	663.9	775.5	
FCFF (DCF engine)	427.1	487.2	568.6	663.9	775.5	

Appendix D - Scenarios and sensitivities (every number is a full model re-run)

	Rev 27E	Rev 28E	EBITDA m 28E	EPS 28E	FCFF 28E	EV	Value/sh	TP	vs CMP	P/E 28E at TP
BSE | bear	4925	5342	63%	67.2	2099	30,385	941	1046	-70.1%	15.6
BSE | base	5578	6220	65%	79.8	2558	47,098	1347	1503	-57.1%	18.8
BSE | bull	5905	6704	66%	86.7	2811	60,405	1670	1869	-46.6%	21.6
BSE | deriv_only	5018	5594	64%	70.8	2230	41,036	1200	1339	-61.8%	18.9
BSE | equity_only	5485	6117	65%	78.3	2504	46,113	1323	1477	-57.8%	18.9
CDSL | bear	1104	1199	44%	19.1	286	3,475	215	237	-81.5%	12.4
CDSL | base	1172	1308	48%	22.7	352	5,698	321	358	-72.1%	15.8
CDSL | bull	1207	1371	50%	24.7	390	7,487	407	456	-64.4%	18.4
CDSL | deriv_only	1168	1304	48%	22.5	349	5,655	319	356	-72.2%	15.8
CDSL | equity_only	1107	1236	45%	20.3	307	5,009	288	321	-74.9%	15.8
CAMS | bear	1486	1637	44%	20.0	406	7,165	312	340	-51.2%	17.0
CAMS | base	1580	1788	47%	23.6	487	11,600	490	539	-22.6%	22.8
CAMS | bull	1627	1874	49%	25.7	533	15,273	637	705	+1.2%	27.5
CAMS | deriv_only	1580	1788	47%	23.6	487	11,600	490	539	-22.6%	22.8
CAMS | equity_only	1486	1682	45%	21.1	428	10,306	438	482	-30.8%	22.9
KFINTECH | bear	1459	1618	35%	21.8	298	5,204	281	306	-64.4%	14.1
KFINTECH | base	1543	1757	40%	26.8	374	9,117	506	560	-34.9%	20.9
KFINTECH | bull	1585	1836	42%	29.7	417	12,366	693	771	-10.3%	26.0
KFINTECH | deriv_only	1543	1757	40%	26.8	374	9,117	506	560	-34.9%	20.9
KFINTECH | equity_only	1459	1661	37%	23.3	318	7,950	439	486	-43.5%	20.8

Bear = derivatives turnover -30% and equity market -20% (persistent level shifts from Q3 FY27), growth -3pp FY28-31, g -1pp. deriv_only / equity_only isolate each brief-mandated shock. Translation path: segment revenue x elasticity -> revenue; fixed costs do not flex -> margin; EPS and FCFF follow; Ke unchanged except where stated; TP = model output.

D1. CAMS: ERP x beta (rf paired with each ERP method)

beta \ ERP	ERP 5.23%	ERP 6.00%	ERP 7.08%
0.94	697	613	633
1.09	608	535	537
1.24	539	475	468
1.39	486	428	415
1.54	442	391	374

D2. CAMS: business driver x market shock

growth \ mkt shock	-0.3	-0.2	-0.1	0.0	0.1
0.07	308	329	350	372	393
0.1	374	399	423	448	473
0.13	454	482	511	539	568
0.16	549	582	615	649	682
0.19	663	701	740	779	818

D3. CAMS: scenario-weight sensitivity of the probability-weighted TP

	w bear	w base	w bull	PW TP
0	0.15	0.60	0.25	551
1	0.25	0.50	0.25	531
2	0.35	0.40	0.25	511
3	0.50	0.25	0.25	481

D4. Tornado (12-month TP, INR) for all four

	TP low	TP high	Range
CAMS | Growth FY28-31 -/+3pp	416	690	274
CAMS | Beta +/-0.2	470	634	164
CAMS | Ke +/-100bp	473	630	157
CAMS | Fixed-cost growth 13%/7%	492	583	92
KFINTECH | Growth FY28-31 -/+3pp	398	760	362
KFINTECH | Beta +/-0.2	482	665	183
KFINTECH | Ke +/-100bp	485	660	175
KFINTECH | Fixed-cost growth 13%/7%	475	639	164
BSE | Growth FY28-31 -/+3pp	1268	1789	521
BSE | Beta +/-0.2	1354	1704	350
BSE | Ke +/-100bp	1360	1693	334
BSE | Derivatives turnover -30%/+15%	1339	1585	246
CDSL | Growth FY28-31 -/+3pp	281	453	173
CDSL | Fixed-cost growth 13%/7%	322	392	70
CDSL | Beta +/-0.2	327	397	70
CDSL | Ke +/-100bp	329	396	67

Appendix E - Deployment timeline, fallback, liquidity; reproduction and sources

	INR cr	Band	Fall needed	P 3m (model)	3m (hist)	P 6m (model)	6m (hist)	P 12m (model)	12m (hist)	Days to fill
T1 starter	100	637	-9%	55%	49%	67%	60%	76%	70%	4.6
T2	150	564	-19%	16%	8%	32%	23%	48%	34%	6.9
T3	250	490	-30%	2%	3%	10%	7%	24%	9%	11.5

Expected INR cr deployed (sum of tranche x touch probability): model 83 / 139 / 208 at 3 / 6 / 12 months; history 68 / 111 / 145. Model = driftless log-normal barrier probability, sigma 30.0% from the last 252 sessions of CAMS.NS; history = all start dates 2021-11-01 to 2026-10-29 (overlapping windows, so not independent). T1 also waits for Q2 FY27 results. Liquidity: average daily traded value INR 109 cr (median 85 cr) over 60 sessions on CAMS.NS+CAMS.BO; at 20% participation INR 500 cr takes ~23 sessions; ticket = 2.9% of market cap.

Timeline and fallback. Month 0-1: no purchase above the T1 band; wait for Q2 FY27 results (Nov-2026). Months 1-6: tranches fill only when bands trade, at most 20% of daily value. Month 6 (after Q4 FY27 results): re-run run_all.py on actuals; bands move only through the model. Month 12: capital still undeployed is returned to the fund's allocation, not forced into the stock and not parked in a sector index (every listed peer is above model value). Until then it sits in overnight/liquid funds. Exit/revisit triggers are in memo Section 5.

Reproduction

`./run_app.sh` (creates .venv from pinned requirements.txt, runs run_all.py, opens the offline Streamlit app) or `make all && make test`. Offline from data/raw; no API keys; optional FIRECRAWL_API_KEY only for refresh.

Sources (ID - description - accessed)

SCR_BSE: Screener.in consolidated page BSE (P&L;/BS/CF/ratios/shareholding/top ratios incl. CMP) (2026-10-29) | **SCR_CDSL**: Screener.in consolidated page CDSL (2026-10-29) | **SCR_CAMS**: Screener.in consolidated page CAMS (2026-10-29) | **SCR_KFIN**: Screener.in consolidated page KFINTECH (2026-10-29) | **SCR_SCHED**: Screener.in schedule API (balance-sheet / P&L; / cash-flow breakdowns) for the four companies (2026-10-29) | **YF**: Yahoo Finance chart API 3y weekly prices for BSE.NS CDSL.NS CAMS.NS KFINTECH.NS ^NSEI (CMP + beta) (2026-10-29) | **BSE_AR26**: BSE Ltd Annual Report FY2025-26 (audited consolidated statements) (2026-10-29) | **BSE_PPT_Q1**: BSE Q1 FY27 investor presentation (2026-10-29) | **BSE_TR_Q1**: BSE Q1 FY27 earnings call transcript (2026-10-29) | **CDSL_AR26**: CDSL Annual Report FY2025-26 (2026-10-29) | **CDSL_PPT_Q1**: CDSL Q1 FY27 investor presentation (slide 18 consolidated income split read from rendered image) (2026-10-29) | **CDSL_TR_Q1**: CDSL Q1 FY27 call transcript (2026-10-29) | **CAMS_AR26**: CAMS Annual Report FY2025-26 (audited consolidated BS/P&L;/CF; notes 5/10/12/18/30/36) (2026-10-29) | **CAMS_PPT_Q1**: CAMS Q1 FY27 investor presentation (2026-10-29) | **CAMS_TR_Q1**: CAMS Q1 FY27 call transcript (2026-10-29) | **KFIN_PPT_Q2**: KFin Q2 FY27 investor presentation incl. Sep-26 statement of assets & liabilities and H1 cash flow (2026-10-29) | **KFIN_PR_Q2**: KFin Q2 FY27 press release (2026-10-29) | **KFIN_TR_Q1**: KFin Q1 FY27 call transcript (2026-10-29) | **KFIN_ASCENT**: KFin intimation: 51% Ascent Fund Services acquisition; remaining 49% in three tranches after FY28/FY29/FY30 linked to EBITDA (2026-10-29) | **HSIE_BSE**: HDFC Securities BSE update 7-Sep-2026 (Q2FY27 QTD premium ADTV ~INR225bn; CAS impact) (2026-10-29) | **SEBI_AR26**: SEBI Annual Report 2025-26 Ch.4 (F&O; measures; index options contracts -52% FY26) (2026-10-29) | **FOW_FO**: FOW: options volumes on NSE/BSE decline >50% in FY26 after SEBI crackdown (2026-10-29) | **BS_STT**: Business Standard: FY26 volumes; Budget-2026 STT hike effective 1-Apr-2026 (2026-10-29) | **MC_RBI**: Moneycontrol: F&O; volumes cool 20-25% as RBI bank-guarantee/prop-funding rules (1-Jul-2026) bite (2026-10-29) | **MINT_CAS**: Mint: BSE and SEBI closing auction session (CAS) impact; options ~95% of transaction charges (2026-10-29) | **CAFE_TER**: Cafemutual: SEBI TER->BER framework effective 1-Apr-2026 (2026-10-29) | **GSEC**: India 10Y G-sec yield 7.40% on 29-Oct-2026 (2026-10-29) | **DAMO_ERP**: Damodaran Jan-2026 country risk premiums: India CDS-based ERP 5.23% (rating-based total ERP 7.08%) (2026-10-29) | **DAMO_RAW**: Damodaran ctryprem.html cached copy (updated Jan 5 2026) (2026-10-29) | **KFIN_SHP**: KFin investor site shareholding pattern 30-Sep-2026 (promoters 14.1%); HBL/PTI 21-Aug-2026 General Atlantic sold 8.75% at avg INR925 -> 13.14% (2026-10-29) | **BSE_CEO**: ET 24-Oct-2026 / Moneycontrol: BSE MD&CEO; Sundararaman Ramamurthy retires 15-Jun-2027 (age 65); applications due 19-Nov-2026 (2026-10-29) | **DAMO_BETA_IN**: Damodaran India industry betas (betaIndia.xls, Industry Averages sheet, updated 5-Jan-2026): unlevered cash-corrected betas used only in the Ke cross- (2026-10-29) | **YF_DAILY**: Yahoo Finance chart API 5y daily OHLCV for CAMS.NS, CAMS.BO, KFINTECH.NS, BSE.NS, CDSL.NS (split-adjusted; volatility, touch frequencies, ADV) (2026-10-29) | **SEBI_CIRC_132**: SEBI circular SEBI/HO/MRD/TPD-1/P/CIR/2024/132 (1-Oct-2024): measures to strengthen equity index derivatives framework (6 measures, effective 20-Nov-2 (2026-10-29) | **SEBI_CIRC_92**: SEBI circular SEBI/HO/MRD/TPD-1/P/CIR/2024/92 (1-Jul-2024, effective 1-Oct-2024): charges levied by MIIs - true to label, uniform non-slab charges (2026-10-29) | **SEBI_PR39**: SEBI PR 39/2025 (7-Jul-2025) EDS vs cash study: ~91% of individual F&O; traders lost in FY25; net losses INR 1,05,603 cr; unique traders 61.4 lakh Q1 - (2026-10-29)